

Definition

- For convex *entropy* function f , we define the f -divergence:

$$D_f(\alpha || \beta) = \int_X f\left(\frac{d\alpha}{d\beta}(x)\right) d\beta(x).$$

- Examples: KL, TV, Hellinger, χ^2 , Jensen Shannon
- Entropy function f determines how to score the ‘discrepancy’ ratio

Examples of f-Divergences